THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A length of 3 km in metres is: 1

(a) 300 m **(b)** 3000 m **(c)** 30 m **(d)** 30000 m

2. The SI unit of length is the: 1

(a) metre **(b)** centimetre **(c)** foot **(d)** kilometre

3. Meghnad Saha explained the spectra of stars by treating stellar matter as: 1

(a) unknowable
(b) ionised gas obeying the laws of physics and chemistry
(c) a liquid
(d) a solid ball

4. A time of 2 hours in seconds is: 1

(a) 720 s **(b)** 3600 s **(c)** 7200 s **(d)** 120 s

5. The SI unit of time is the: 1

(a) day **(b)** second **(c)** minute **(d)** hour

6. Mathematics helps science mainly by: 1

(a) avoiding experiments
(b) describing patterns precisely and making predictions
(c) making science harder
(d) replacing observations

7. A speed of 36 km/h in m/s is: 1

(a) 100 m/s **(b)** 10 m/s **(c)** 3.6 m/s **(d)** 36 m/s

8. The symbol c is used for the speed of light because: 1

(a) it is a random choice
(b) light has no speed
(c) it is the first letter of 'cm'
(d) scientists agree on common symbols to communicate clearly

9. Measurement of the same length by two students gives 15.2 cm and 15.3 cm. This shows: 1

(a) one student is wrong **(b)** length has no meaning
(c) measurements have some uncertainty **(d)** the scale is faulty always

10. Which of the following needs an exact value rather than an estimate? 1

(a) the size of a crowd **(b)** the age of a mountain
(c) the dose of a medicine **(d)** the number of stars

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State why 'no scientific theory is final' is a strength of science. 2

12. What is a prediction? How is it different from a guess? 2

13. Convert 5 km/h into m/s. 2

14. What is meant by chemistry? Name two areas it influences. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A friend says 'a theory is just a guess'. Respond in three points. 3

16. State three reasons why science values careful reasoning. 3

17. Estimate the number of seconds in a day and a year (365 days). 3

OR

Convert: 3
(a) 72 km/h to m/s
(b) 5 m/s to km/h.

18. Name three types of chemical changes seen in daily life and one sign for each. 3

OR

How does chemistry help to improve daily life? Give three examples. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Distinguish between a law, a hypothesis and a theory with two examples and explain how a theory is tested. 5

OR

Explain the importance of unit consistency using the aircraft incident and three conversion examples. 5

20. Explain Meghnad Saha's contribution to astrophysics and the role of mathematics in science. 5

OR

Describe how weather forecasting shows the strengths and limits of models. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND